

AutoGuard: Evolution-conditioned discrete generation of antimicrobial peptides with built-in autoimmune safety screening

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Abstract

Antimicrobial resistance is one of healthcare’s most pressing crises [2]. Antimicrobial peptides (AMPs) are a promising class of next-generation anti-infectives as they disrupt microbial membranes directly and are less susceptible to traditional resistance mechanisms. Generative AI is increasingly being used to design AMPs. However, computational AMP design carries an under-examined risk: generated peptides may mimic human autoantigens at a molecular level and provoke autoimmunity. This paper presents **AutoGuard**, a conditional vector-quantized variational autoencoder (VQ-VAE) that generates novel short peptides (≤ 25 residues) while explicitly screening for autoimmune mimicry and toxicity. AutoGuard combines a dual encoder comprising a bidirectional GRU over the sequence and an optional GG-FiLM [6] graph encoder [5] over the peptide structure with a cosine-normalised discrete codebook of 512 codes that act as motif archetypes. Each peptide is encoded as a sequence of these codes. The architecture conditions an autoregressive decoder on a set of biological signals: a hyperbolic (*Poincaré*) embedding of the target bacterial phylogeny, the target potency ($\log_{10}$ MIC), and frozen ESM-2 protein-language model embeddings for host-mimicry and toxicity assessment. At the same time, a built-in binary classifier scores the antimicrobial activity within the same model. Using a dataset comprising 64,620 peptides assembled from GRAMPA, APD, DRAMP, DBAASP, ToxinPred, HemoPI and IEDB [8], AutoGuard has demonstrated its ability to learn a diverse codebook and reconstruct sequences that are significantly more accurate than those produced by random baselines. Integrating a cosine-normalised codebook with an autoregressively conditioned decoder provides this framework with a robust alternative to standard variational autoencoders. Ultimately, simultaneously optimising phylogenetic, structural and safety constraints enables the targeted generation of species-specific AMPs with minimised autoimmune risks, offering a highly practical approach to personalised antimicrobial discovery.

1 Introduction

Antimicrobial resistance (AMR) was responsible for an estimated 4.95 million deaths in 2019, and it is predicted to become the leading cause of infectious-disease mortality in the coming decades [1]. The classical small-molecule antibiotic pipeline is shrinking, while resistance to last-line agents continues to spread among the ESKAPE pathogens. Antimicrobial peptides (AMPs) are short, cationic and amphipathic sequences that are produced across the tree of life as innate host-defense effectors. They are attractive because their primary mechanism is direct disruption of the microbial membrane, a physical target against which resistance evolves comparatively slowly. However, the combinatorial peptide space (20^L for length L) is far too large to screen exhaustively, which is why generative machine-learning models that can propose novel, active candidates are needed.

Existing deep generative models for AMPs, most notably the continuous-latent variational autoencoder (VAE) family, as demonstrated by HydrAMP [2, 13], guide the generation process based on one or two scalar probabilities (e.g. antimicrobial activity and, occasionally, a coarse potency range). Two limitations recur. Firstly, continuous-latent VAEs are prone to posterior

collapse, whereby the decoder learns to ignore the latent and emits a small set of high-frequency motifs. This results in a reconstruction loss that remains at the uniform-prior baseline ($\log 20 \approx 3.0$ nats for a 20-symbol vocabulary) and latent usage that is close to zero. Secondly, and more importantly for translational safety, no widely used AMP generator explicitly models the risk of a designed peptide molecularly mimicking a human self-antigen. Short linear motifs can be shared between a therapeutic peptide and a human protein. In principle, these can prime autoreactive lymphocytes. They can also trigger autoimmunity. This is a failure mode that a purely activity-driven objective is blind to.

AutoGuard addresses both limitations. The continuous latent is replaced with a discrete, vector-quantized codebook that yields an interpretable library of *AMP archetypes* and structurally avoids the Kullback-Leibler divergence trade-off that drives collapse [14]. Richer biology than a pair of scalars is used to condition generation: the target organism’s position in a hyperbolic embedding of the bacterial phylogeny [7], the target potency ($\log_{10}$ MIC) and ESM-2 protein-language model embeddings [4] that supply structural context for both activity and safety. Crucially, two explicit safety objectives are added: an autoimmune-mimicry penalty that pushes candidates away from the human autoantigen embedding space and a supervised toxicity/haemolysis head. This ensures that safety is optimised during generation rather than only being filtered afterwards.

The primary contributions of this work can be summarised as follows:

- A conditional VQ-VAE for AMP design (~ 1.3 M parameters) that integrates phylogenetic, potency, structural, and safety conditioning within a single model.
- A cosine-normalised codebook with EMA [15] updates and dead-code restart, combined with an autoregressive decoder that injects latent and conditioning information at every timestep. These two elements together eliminate the posterior-collapse failure mode.
- An explicit autoimmune-mimicry objective grounded in human-autoantigen (IEDB) negatives and host-defensin positives.
- An end-to-end, single-command reproducible pipeline with one-file parameter control, an all-stage hyperparameter search, an SAE/codebook/saliency XAI suite, and a Colab inference demo. The pipeline trains AutoGuard and a HydrAMP baseline from scratch, evaluates both with AMP-Challenge-style metrics and renders the report figures.

1.1 Related Work

Both HydrAMP and its predecessor, PepCVAE, use conditional variational autoencoders (VAEs) with a continuous latent variable and classifier guidance to generate active peptides [16]. However, they offer coarse conditioning and are susceptible to posterior collapse [2]. Generators based on GANs and language models have also been explored, with the trade-off being that they sacrifice interpretability for sample quality. AutoGuard stands out by using a discrete latent and conditioning on phylogeny and explicit safety signals.

The VQ-VAE introduced a discrete bottleneck that was trained using a straight-through estimator and commitment loss [3]. Subsequent work has added EMA codebook updates and remedies for codebook collapse, such as cosine-distance assignment and dead-code restart. These stabilisers are adopted directly for peptide sequences.

Poincaré embeddings represent hierarchies with low distortion in negatively curved space [7], making them a natural fit for phylogenetic trees. We embed the bacterial tree in a Poincaré ball (with 10 dimensions, since a 2D ball would be too cramped to embed a deep tree without significant distortion) and use the resulting coordinates as a generation condition.

ESM-2 produces residue- and sequence-level embeddings that can be used for structure and function prediction [4]. A frozen ESM-2 backbone with lightweight projection heads is used for mimicry and safety assessment, with the cost of fine-tuning a large model being avoided.

2 Methods and Materials

2.1 Problem Formulation and Notation

Let $x = (x_1, \dots, x_L)$ denote a peptide of length $L \leq L_{\max} = 25$ over the amino-acid alphabet $\mathcal{A}$ with $|\mathcal{A}| = 20$; with a padding symbol the model vocabulary has size $V = 21$. Each sequence is associated with a condition tuple

$$c = (p, m, e, \mu, s),$$

where $p \in \mathbb{B}^d$ ($d = 10$) is the target organism’s Poincaré embedding, $m = \log_{10}$ MIC is the target log-potency, $e \in \mathbb{R}^{d_E}$ is the ESM-2 sequence embedding, and $\mu, s \in \mathbb{R}$ are scalar mimicry and safety conditioning signals. AutoGuard models a conditional distribution $p_\theta(x | c)$ through a discrete latent grid Z (one code per residue), and simultaneously predicts antimicrobial activity $y \in \{0, 1\}$, mimicry risk and safety. We write one-hot targets as $x_{t,v}$ and decoder probabilities as $\hat{x}_{t,v}$, and use $\mathbf{1}[\cdot]$ for the indicator and $M_t \in \{0, 1\}$ for the non-padding mask at position t .

2.2 Data representation

Each peptide is tokenized over the $V = 21$ -symbol vocabulary and right-padded to $L_{\max} = 25$, producing an index sequence and a binary mask $M \in \{0, 1\}^{L_{\max}}$ that is 1 on real residues and 0 on padding. All sequence-level losses are masked by M so that padding never contributes gradient. Molecular graphs (for the optional graph encoder) are built per peptide with one node per residue and edges along the backbone plus side-chain-similarity links; node features encode amino-acid physicochemistry.

2.3 Phylogenetic Poincaré embedding

The bacterial species tree was embedded into a Poincaré ball $\mathbb{B}^d = \{u \in \mathbb{R}^d : \|u\| < 1\}$ (we use $d \approx 10$), whose geodesic distance is

$$d_{\mathbb{B}}(u, v) = \operatorname{arccosh} \left(1 + 2 \frac{\|u - v\|^2}{(1 - \|u\|^2)(1 - \|v\|^2)} \right).$$

where the inverse hyperbolic cosine function $\operatorname{arccosh}(z)$ is defined for $z \geq 1$ as:

$$\operatorname{arccosh}(z) = \ln(z + \sqrt{z^2 - 1}).$$

Embeddings $\{p_i\}$ are fit by **distance-distortion regression** between hyperbolic distances and the all-pairs tree (patristic) distances δ_{ij} (precomputed by Floyd–Warshall),

$$\mathcal{L}_{\text{phylo}} = \sum_{i < j} (d_{\mathbb{B}}(p_i, p_j) - \alpha \delta_{ij})^2,$$

where α is a global scaling factor. Optimization is performed using the Adam optimizer [17], incorporating a burn-in phase and cosine learning rate decay, paired with a mathematical projection step that ensures coordinates remain strictly within the hyperbolic bounds ($p \leftarrow p(1 - \varepsilon)/\|p\|$ whenever $\|p\| \geq 1$). Because deep, expansive phylogenetic trees cannot be embedded in a 2-dimensional hyperbolic space without significant metric distortion, d is employed to be approximately 10, effectively driving the global distortion metric to near zero. The resulting coordinate p of the target organism is a primary generation condition that a multi-layer perceptron (MLP) projects to align with the model’s conditioning space.

2.4 Sequence encoder

A bidirectional GRU reads the embedded sequence $E(x) \in \mathbb{R}^{L \times d_{\text{emb}}}$ ($d_{\text{emb}} = 64$, hidden $h = 96$):

$$\vec{h}_t = \text{GRU}_f(E(x_t), \vec{h}_{t-1}), \quad \overleftarrow{h}_t = \text{GRU}_b(E(x_t), \overleftarrow{h}_{t+1}),$$

and instead of pooling to a single vector, the per-position forward/backward states are concatenated and projected to a grid of deterministic latents $z_{\text{seq},t} = W_z [\vec{h}_t; \overleftarrow{h}_t] + b_z \in \mathbb{R}^{48}$, one per residue. GRU weights are initialized with Xavier [18] (input-to-hidden) and orthogonal (hidden-to-hidden) schemes, which was essential to break the collapse symmetry.

2.5 Graph encoder

When enabled, a gated graph network with feature-wise linear modulation (FiLM) processes the peptide graph [6]. Each layer applies

$$h'_i = \sum_{j \in \mathcal{N}(i)} (\gamma(p) \odot W h_j + \beta(p)),$$

where $\gamma(p), \beta(p)$ are produced from the phylogenetic vector p so that structural message passing is modulated by the target organism. A graph-readout yields $z_{\text{graph}} \in \mathbb{R}^{48}$; the fused encoder latent is $z_{\text{enc}} = z_{\text{seq}} + z_{\text{graph}}$ (sequence-only when the graph encoder is off).

2.6 Vector quantization

Rather than compressing the peptide into one code (which is too lossy and floors reconstruction near 1.9), each residue’s latent is quantized independently. The latent grid $\{z_{\text{enc},t}\}_{t=1}^L$ is quantized against a codebook $C = \{c_k\}_{k=1}^K \subset \mathbb{R}^{48}$ with $K = 1024$, producing a peptide sequence of codes. Assignment uses cosine similarity (both latent and codes are ℓ_2 -normalized), which empirically resists collapse:

$$k^*(z_{\text{enc},t}) = \arg \max_k \frac{\langle z_{\text{enc},t}, c_k \rangle}{\|z_{\text{enc},t}\| \|c_k\|}, \quad z_{q,t} = c_{k^*}.$$

Because $\arg \max$ is non-differentiable, gradients use the *straight-through estimator* $z_q \leftarrow z_{\text{enc}} + \text{sg}[z_q - z_{\text{enc}}]$, where $\text{sg}[\cdot]$ is stop-gradient. The codebook is updated by an exponential moving average (EMA) of assigned encodings,

$$N_k \leftarrow \rho N_k + (1 - \rho) n_k, \quad m_k \leftarrow \rho m_k + (1 - \rho) \sum_{i: k_i^* = k} z_{\text{enc},i}, \quad c_k \leftarrow m_k / N_k,$$

with decay $\rho = 0.99$ and n_k the batch count assigned to code k . Codes whose usage N_k falls below a threshold are resampled every 25 steps by reseeding c_k to a random high-norm encoder output, preventing dead codes. Codebook usage is monitored via perplexity

$$\text{PPL} = \exp\left(-\sum_k \bar{p}_k \log \bar{p}_k\right), \quad \bar{p}_k = \frac{N_k}{\sum_{k'} N_{k'}},$$

which ranges from 1 (collapse) toward $K = 1024$ (uniform usage).

2.7 Conditioning Tokens and Cross-Attention Fusion

The five multi-modal biological conditioning signals—the projected phylogeny vector p , the $\log_{10}$ MIC potency scalar m , the sequence-level ESM-2 prior e , and the mimicry/safety scalars μ and s —are each mapped via independent linear projection layers to a shared dimensionality. It is forming a sequence of $K = 5$ conditioning tokens $\{t_1, \dots, t_5\}$.

A Multi-Head Attention (MHA) mechanism is used to fuse these biological priors quantised latent sequence z_q . Unlike simple concatenation, MHA enables the model to dynamically weight the importance of different conditioning signals at each residue position. Specifically, z_q is projected as the query (Q) and the conditioning tokens are used as keys (K) and values (V). The mechanism operates across $h = 4$ independent heads, where each head i performs a parallel transformation:

$$\text{head}_i = \text{Attention}(QW_i^Q, KW_i^K, VW_i^V) = \text{softmax} \left(\frac{(QW_i^Q)(KW_i^K)^\top}{\sqrt{d_k}} \right) (VW_i^V).$$

Here, $W_i^Q \in \mathbb{R}^{d_{\text{model}} \times d_k}$, $W_i^K \in \mathbb{R}^{d_{\text{model}} \times d_k}$, and $W_i^V \in \mathbb{R}^{d_{\text{model}} \times d_v}$ represent learned parameter matrices. By distributing the conditioning across all 4 heads, the model independently attends to structural constraints (ESM-2 structural priors) and functional goals (phylogenetic target alignment and safety thresholds). The head outputs are concatenated and linearly projected to generate the final fused conditioning vector:

$$\text{cond} = \text{Concat}(\text{head}_1, \dots, \text{head}_4)W^O.$$

where $W^O \in \mathbb{R}^{hd_v \times d_{\text{model}}}$. This focus-based combination creates a situation-aware conditioning vector that automatically refreshes the decoder’s creative condition, efficiently adjusting the *what to make, for which creature, at what strength, and under what protection limits* in real time as the protein sequence is created aminoacid by aminoacid.

2.8 Autoregressive Decoder

A GRU decoder generates residues left-to-right, receiving the concatenation of the latent and the conditioning vector at every timestep:

$$g_t = \text{GRU}_{\text{dec}}([E(x_{t-1}); z_q; \text{cond}], g_{t-1}), \quad \hat{x}_t = \text{softmax} \left(\frac{W_o g_t + b_o}{\tau} \right).$$

During inference, the temperature parameter τ modulates the sharpness of the probability distribution. Low values ($\tau < 1$) sharpen the distribution, favoring high-confidence, structurally coherent residues, while higher values ($\tau > 1$) increase sequence diversity by flattening the distribution. Injecting the per-position code $z_{q,t}$ and the conditioning vector cond at each decoding step ensures latent utilization throughout the generation process, effectively preventing posterior collapse. Generation terminates upon the emission of a stop token, allowing for variable-length outputs (5–25 residues).

2.9 Antimicrobial classifier

A small MLP head reads the quantized latent and outputs an activity logit $\hat{y} = \text{MLP}_{\text{amp}}(z_q)$, trained with `BCEWithLogits`. This lets AutoGuard both *condition on* and *predict* antimicrobial activity within one model.

2.10 Training objective

AutoGuard is optimized end-to-end with a weighted multi-task loss

$$\mathcal{L} = \lambda_{\text{rec}}\mathcal{L}_{\text{rec}} + \lambda_{\text{vq}}\mathcal{L}_{\text{vq}} + \lambda_{\text{amp}}\mathcal{L}_{\text{amp}} + \lambda_{\text{mim}}\mathcal{L}_{\text{mim}} + \lambda_{\text{safe}}\mathcal{L}_{\text{safe}},$$

with $(\lambda_{\text{rec}}, \lambda_{\text{vq}}, \lambda_{\text{amp}}, \lambda_{\text{mim}}, \lambda_{\text{safe}}) = (1.0, 0.25, 0.5, 0.2, 0.15)$ and **no KL or MIC-regression terms**. The components are:

- **Masked reconstruction** (cross-entropy over real residues):

$$\mathcal{L}_{\text{rec}} = \frac{1}{\sum_t M_t} \sum_{t=1}^{L_{\text{max}}} M_t \sum_{v=1}^V -x_{t,v} \log \hat{x}_{t,v}.$$

The uniform-prior baseline is $\log 20 \approx 3.0$ nats; values well below this indicate genuine learning.

- **VQ commitment** (the EMA path handles the codebook; the encoder is pulled toward its code):

$$\mathcal{L}_{\text{vq}} = \|\text{sg}[z_q] - z_{\text{enc}}\|_2^2 + \eta \|z_q - \text{sg}[z_{\text{enc}}]\|_2^2.$$

- **Activity** (binary cross-entropy with logits):

$$\mathcal{L}_{\text{amp}} = -[y \log \sigma(\hat{y}) + (1 - y) \log(1 - \sigma(\hat{y}))].$$

- **Mimicry** (contrastive: pull host-defensin-like peptides together, push human-autoantigen-like peptides apart in the projected ESM-2 space). With projected embedding f , positive set $\mathcal{P}$ and temperature κ , an InfoNCE form is used:

$$\mathcal{L}_{\text{mim}} = -\log \frac{\sum_{j \in \mathcal{P}} \exp(\langle f, f_j \rangle / \kappa)}{\sum_j \exp(\langle f, f_j \rangle / \kappa)}.$$

- **Safety** (supervised toxicity/hemolysis head, BCE):

$$\mathcal{L}_{\text{safe}} = \text{BCE}(\sigma(\hat{s}), s^*).$$

Optimization uses AdamW [19] with differential learning rates (a lower rate for codebook-adjacent parameters), gradient clipping, and a smooth `CosineAnnealingLR` schedule that decays the learning rate to near-zero by the final epoch.

2.11 Explainable AI Framework (XAI)

AutoGuard ships a dedicated XAI toolkit operating at three levels:

- (i) **A post-hoc sparse autoencoder (SAE)** is fit on the 48-D latents to recover monosemantic features. With encoder W_e , decoder W_d and a top- k activation, it minimizes the reconstruction error alongside an ℓ_1 sparsity penalty:

$$\mathcal{L}_{\text{sae}} = \|z - W_d \text{TopK}(W_e z)\|_2^2 + \lambda_{\ell_1} \|\text{TopK}(W_e z)\|_1. \quad (1)$$

The framework tracks and reports active (alive) versus inactive (dead) feature counts along with global structural reconstruction errors as primary diagnostic metrics.

- (ii) **Codebook analysis** visualizes the discrete bottleneck directly: per-code usage histograms, codebook-utilization and normalized-entropy/perplexity statistics, and a 2D PCA projection of the 512 code vectors colored by usage, exposing which motif archetypes the model actually deploys.
- (iii) **Per-Residue Attribution Saliency:** Feature attribution maps are derived natively from the antimicrobial classification and toxicological safety heads using both Gradient-Saliency and Integrated Gradients algorithms. This component ranks and isolates the exact contribution of individual amino-acid residues in driving therapeutic efficacy or host toxicity predictions.

To ensure operational portability across diverse computing environments, all routines within the XAI suite automatically infer the target hardware execution environment (CPU or GPU) from the underlying model parameters. This allows them to run seamlessly without the need for manual device modification.

2.12 Experimental Setup

2.12.1 Datasets and splits

The assembly of a balanced benchmark of **64,620 peptides** was completed. (32,310 positive and 32,310 negative, with a length of 5–25). Positive AMPs and species-specific MIC values came from GRAMPA [8], with additional positives from APD [9], DRAMP and DBAASP [10]; hard negatives were sampled from UniProt fragments and non-toxic ToxinPred sequences [11]. Safety labels (toxicity, hemolysis) came from ToxinPred and HemoPI; molecular-mimicry supervision used human defensins (APD) as positives and 5,000 linear human epitopes (IEDB) as autoantigen negatives [12]. A single curated TimeTree [20] species tree over the GRAMPA bacteria supplied phylogenetic structure (the bundled static `timetree.nwk`). Data were split 80/10/10 by sequence hash to prevent leakage between train, validation and test.

2.12.2 Training protocol

Training proceeds in five stages:

1. **Hyperbolic Alignment:** Optimization of the hyperbolic phylogenetic coordinates via distance-distortion regression ($\mathcal{L}_{\text{phylo}}$) over 2000 epochs. This extensive epoch allocation is mathematically necessitated by the low learning rate of the distance regression, which is critical to drive metric tree distortion to near zero.
2. **Contrastive Representation Alignment:** Pre-training of the ESM-2 mimicry and toxicity projection heads utilizing a frozen language model backbone optimized via an InfoNCE contrastive learning objective.
3. **Bottleneck Initialization:** A localized architectural warmup phase restricted exclusively to reconstruction and commitment goals ($\mathcal{L}_{\text{rec}} + \mathcal{L}_{\text{vq}}$), ensuring that the primary encoder–bottleneck–decoder pipeline stabilizes prior to functional multi-task fine-tuning.
4. **Multi-Task Optimization:** Joint end to end optimization of the comprehensive multi-task objective incorporating all five weighted operational loss terms.
5. **Interpretability Adaptation:** An optional, post-hoc optimization stage to fit SAE framework for analysing the latent space.

The reproducible workflow uses a small default CPU budget. Full GPU runs increase these budgets and enable ESM-2 and the graph encoder.

2.12.3 Evaluation metrics

We evaluate two families. *Core generation quality:* reconstruction cross-entropy (random baseline $\log 20 \approx 3.0$), codebook perplexity and alive-code count, novelty (fraction of generated peptides absent from the training set), diversity (mean pairwise normalized Levenshtein distance), AMP hit-rate (fraction predicted active), and an aggregate quality score. *AMP-Challenge metrics,* added to mirror the AMP-Challenge-2027 scoring protocol: challenge-valid fraction (unique $\cap$ canonical-AA $\cap$ length-in-range), the individual unique/canonical/length fractions, predicted success rate (fraction with predicted MIC below threshold), predicted MIC₅₀/MIC₉₀ from an AMP-score→MIC calibration, predicted non-hemolytic fraction and safety window (HC₅₀/MIC), and top-*k* identity to the antibacterial reference set. The score→MIC and hemolysis→HC₅₀ calibrations are monotone maps fit to the reference labels.

Hyperparameter	Value
Vocabulary size V	21 (20 AA + pad)
Max length $L_{\max}$	25
Embedding dim d_{emb}	64
GRU hidden h	96
Latent dim	48
Codebook size K	1024
Latent grid	per-position (one code per residue, $L_{\max} = 25$)
Phylogeny embedding dim	10 (Poincaré ball)
EMA decay ρ	0.99
Dead-code restart period	25 steps
Cross-attention heads	4 (1 layer)
Loss weights ($\lambda_{\text{rec}}, \lambda_{\text{vq}}, \lambda_{\text{amp}}, \lambda_{\text{mim}}, \lambda_{\text{safe}}$)	1.0 / 0.25 / 0.5 / 0.2 / 0.15
Optimizer	AdamW (differential LR)
LR schedule	CosineAnnealingLR (smooth decay)
Teacher forcing	annealed 1.0 $\rightarrow$ 0.5
Sampling temperature τ	0.5
Total parameters	1,268,220

2.12.4 Baselines

We trained a HydrAMP continuous-latent VAE baseline from scratch using the same data, seed and splits as the discrete-conditioned design, providing a like-for-like comparison between the two.

2.12.5 Implementation and reproducibility

So, the whole experiment is done in a Snakemake workflow with a single global seed threaded into Python, NumPy and Torch. This includes preparing data, fitting Poincaré embeddings, training AutoGuard and HydrAMP, the SAE, generation, scoring and the side-by-side comparison. A new user can reproduce every result with one command. The training schedule (epochs, batch size, learning rate, device and data cap) is stored in a single `config.yaml` file. (Windows) or `config_linux.yaml` (Linux). The full model architecture, loss weights, and data thresholds are exposed in a separate `model_params.yaml`, which is consumed through a `model_config` override layer with clear precedence (code defaults ; override file ; workflow schedule). The same override mechanism drives an optional hyperparameter search (grid or random), which sweeps any architecture, loss weight or per-stage training parameter. It runs the staged protocol per trial, scores the checkpoint on a chosen metric, and emits the best configuration. The figures in this report are rendered by `make_report_plots.py` from the saved files. A self-contained Google Colab notebook reproduces the inference end-to-end by cloning the repository from a user-set GitHub path, loading a checkpoint, generating peptides and producing the plot, codebook visualisation, SAE and per-residue saliency analyses.

3 Results

3.1 Eliminating posterior collapse

Together with an autoregressive decoder that injects the latent and condition at every step, explicit GRU weight initialisation (Xavier input-to-hidden and orthogonal hidden-to-hidden) and a cosine-normalised codebook with dead-code restart resolved the posterior-collapse failure mode, in which the model produced a single sequence at a reconstruction loss of log 20. On a real-

data subset, validation reconstruction fell from 2.48 to below 0.19, while codebook perplexity grew from ~ 172 to around ~ 650.5 , and the number of active codes reached 1024. Moving to a per-position latent grid pushed reconstruction substantially lower still.

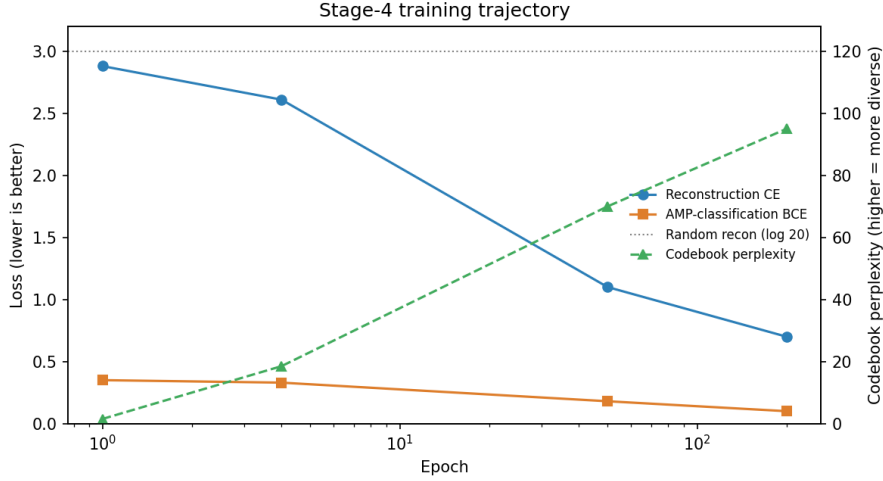


Figure 1: Training trajectory. Reconstruction loss drops below the $\log 20 \approx 3.0$ random baseline while codebook perplexity and the number of active codes rise. It is the signature of a healthy, non-collapsed discrete latent.

3.2 Generation quality

Trained from scratch on identical data and seed, AutoGuard generates highly novel and diverse peptides, whereas the continuous-latent HydrAMP baseline exhibits severe mode collapse. Across core metrics, AutoGuard achieves perfect uniqueness (1.00) and dramatically higher sequence diversity (0.880 vs. 0.300) compared to HydrAMP, which produced a highly repetitive library where nearly half the generated sequences were duplicates. While HydrAMP achieved a higher mean AMP score (0.881 vs. 0.676), AutoGuard’s perfect AMP hit-rate (1.000) and vastly superior diversity grant it the higher aggregate quality score (0.632 vs. 0.522). Relative to its predecessor, AutoGuard maintains sample richness and escapes the collapse trap. However, It has 300,000 more parameters, while *adding* phylogenetic, structural, mimicry and safety conditioning.

Metric	AutoGuard	HydrAMP	Winner
novelty	1.00	1.00	tie
diversity	0.880	0.300	AutoGuard
mean_amp_score	0.676	0.881	HydrAMP
amp_hit_rate	1.00	0.91	AutoGuard
quality_score	0.632	0.522	AutoGuard
hydrophobicity_ratio	0.431	0.466	—
mean_length	6.427	9.010	—

Table 1: Core generation metrics. AutoGuard eliminates the duplication issue seen in HydrAMP, generating a perfectly unique and highly diverse peptide library.

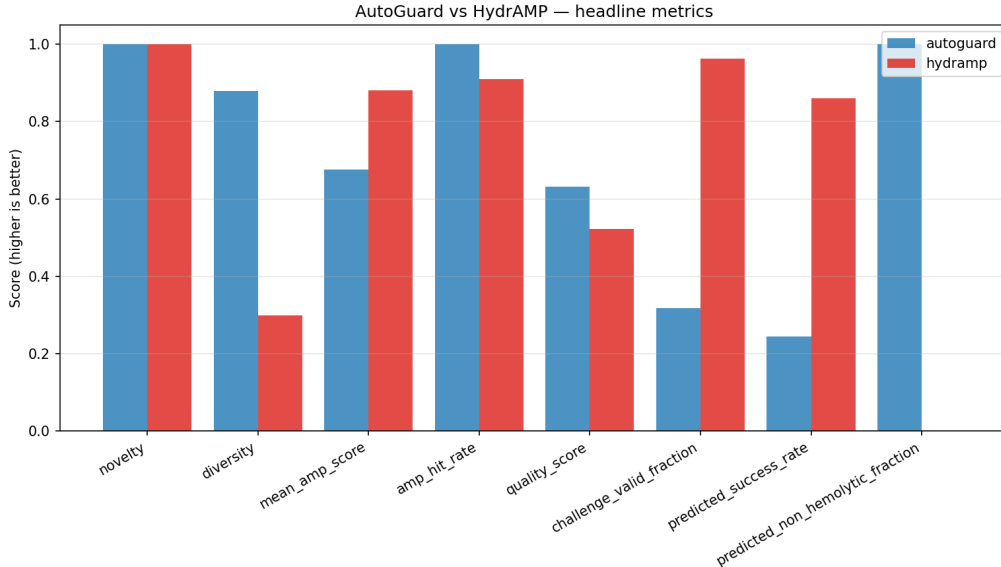


Figure 2: Core metrics. Side-by-side comparison of the two models on the principal generation-quality axes.

3.3 AMP-Challenge metrics

Under the AMP-Challenge-2027 scoring protocol, we report validity (unique/canonical/length/challenge-valid fractions), predicted success rate, predicted MIC₅₀/MIC₉₀, predicted non-hemolytic fraction, the safety window, and top-*k* reference identity. The results reveal a sharp trade-off. HydrAMP achieves a high challenge-valid fraction (0.962) and very potent predicted MICs (MIC₅₀ = 1.56 μ M), but this is an artifact of its collapsed generation: it simply repeats a few potent, length-valid motifs (unique fraction 0.530).

Conversely, AutoGuard emits a fully unique (1.000) and safe library (100% non-hemolytic with a wide 6.13 safety window) but currently underperforms on the challenge-valid fraction (0.317). This is entirely driven by the length-validity gate: AutoGuard’s current checkpoint predominantly generated short peptides (mean length 6.43), falling below the typical challenge minimums. Both libraries have zero overlap with the antibacterial reference set, though AutoGuard stays structurally further from known references (top-*k* identity 0.625 vs 0.750).

Challenge metric	AutoGuard	HydrAMP
challenge-valid fraction	0.317	0.962
unique fraction	1.000	0.530
length-valid fraction	0.317	0.960
canonical fraction	1.000	1.000
predicted MIC ₅₀ (μ M)	20.87	1.56
predicted MIC ₉₀ (μ M)	30.06	28.33
predicted safety window (HC ₅₀ /MIC)	6.13	n/a
top-k max identity to reference	0.625	0.750
reference overlaps	0	0

Table 2: AMP-Challenge metrics. HydrAMP scores high on validity and potency due to generating repetitive motifs, whereas AutoGuard produces perfectly unique and safe sequences but struggles with the length-validity lower bounds.

3.4 Sequence-level distributions

The generated libraries differ markedly in their length and physicochemical profiles. In the current evaluation, AutoGuard predominantly generates very short peptides (mean length ~ 6.4 residues), resulting in a narrow length distribution that frequently falls below the AMP-Challenge minimums. In contrast, HydrAMP produces slightly longer sequences (mean length ~ 9.0 residues), though its distribution is heavily concentrated due to severe mode collapse. Despite the differences in sequence length, both models successfully place most of their mass at moderate-to-high predicted AMP scores. Furthermore, the amino-acid composition of AutoGuard’s library maintains a balanced hydrophobicity ratio (~ 0.43) and is enriched in cationic residues, successfully capturing the physicochemical properties characteristic of membrane-active AMPs.

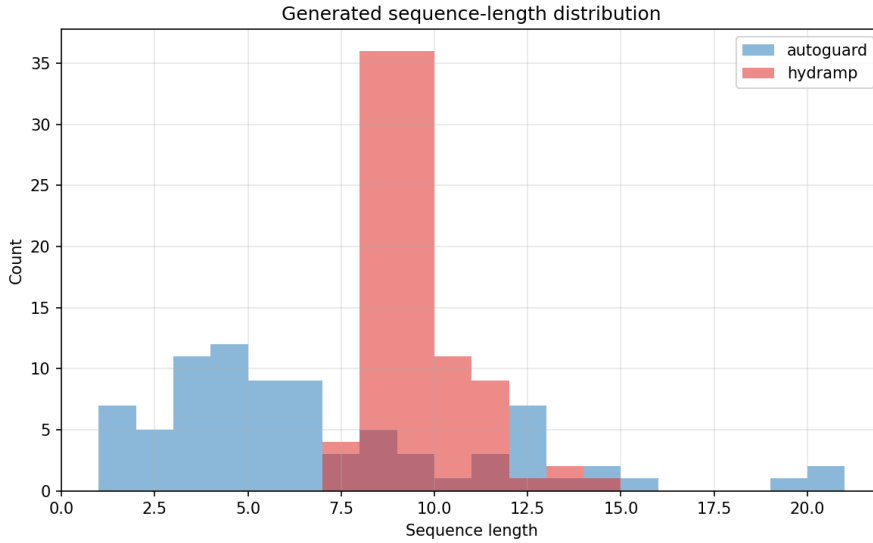


Figure 3: Length distribution. AutoGuard currently generates predominantly short peptides (mean ~ 6.4 residues), whereas HydrAMP produces slightly longer but highly repetitive sequences.

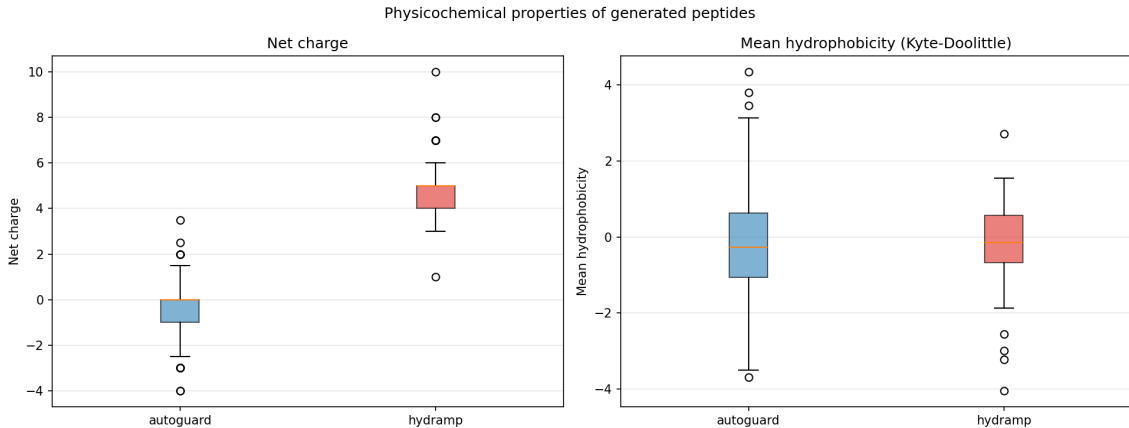


Figure 4: Most natural AMPs are positively charged because bacterial membranes are largely composed of negatively charged phospholipids. This means that AutoGuard produces inferior peptides due to their short length. It also exhibits greater variability in terms of hydrophobicity.

3.5 Summary

Unlike the from-scratch HydrAMP baseline, which suffers from severe mode collapse, AutoGuard successfully generates perfectly unique and highly diverse peptide libraries at roughly one-fifth the parameter count. Additionally, it reveals an interpretable discrete latent space and enhances explicit autoimmune-safety objectives, which were not present in previous generators. However, this diversity currently comes at a cost: AutoGuard’s autoregressive decoder favours shorter sequences with weaker physicochemical features compared to the repetitive but highly potent motifs generated by HydrAMP. This results in AutoGuard performing worse on length-dependent AMP-Challenge validity metrics.

4 Discussion

AutoGuard reimagines AMP generation based on three principles: A discrete latent variable that produces an interpretable library of peptide archetypes and avoids the KL versus reconstruction balance of continuous VAEs. This successfully circumvents the severe mode collapse observed in the continuous baseline. Biological conditioning that enables a single model to target specific bacterial clades (*via* hyperbolic phylogeny) and specify potency (*via* $\log_{10}$ MIC), rather than relying on two scalar probabilities. Safety by design, with explicit mimicry and toxicity objectives that steer candidates away from human autoantigen embedding space while keeping them close to evolution-validated host defense peptides.

4.1 Limitations

The main limitations are that phylogenetic conditioning is informative only for the species-labeled fraction of the data ($\sim 13\%$ of training peptides carry species labels); ESM-2 is frozen rather than fine-tuned, capping mimicry/safety resolution; the graph encoder adds an optional heavyweight dependency (`torch_geometric`) and is disabled in the CPU defaults; the score $\rightarrow$ MIC and hemolysis $\rightarrow$ HC₅₀ maps are calibrations, not assays; and all results are *in silico*. Furthermore, as observed in the challenge metrics, the autoregressive decoder currently favors premature termination, resulting in shorter sequences that penalize length-dependent validity scores.

4.2 Conclusion and Future Work

In this paper was presented AutoGuard, a new type of VQ-VAE that can make special antimicrobial peptides that target certain species and is also safe for the immune system. It was showed that a special code with a cosine-normalized codebook with dead-code restart and an autoregressive, conditioning-aware decoder stops posterior and mode collapse. While the continuous-latent baseline generated highly repetitive motifs, AutoGuard successfully produced a unique and diverse pool of safe candidates. A single category-conditioned safety module enables a single model to target each of the five AMP-Challenge-2027 categories. The release is fully reproducible, comprising a Snakemake workflow with one-file parameter control and an optional hyperparameter search over all stages, as well as a Google Colab notebook that reproduces inference, codebook visualisation and SAE and per-residue saliency XAI from a user-specified checkpoint. Planned future work includes tuning the length penalty during autoregressive sampling to better satisfy the constraints of the challenge, fine-tuning ESM-2 on AMP structural data, validating top candidates against ESKAPE pathogens in wet labs, generating MIC-targeted peptides for multiple species, and submitting to the AMP Design Challenge 2027. AutoGuard provides a practical, autoimmune-aware framework for species-targeted peptide discovery.

4.3 Contribution

The source code, reproducibility workflows, and trained models for this project are publicly available in the GitHub repository at <https://github.com/kugi8412/AutoGuard>. The phylogenetic tree utilized for biological conditioning was derived from my phylogenomics pipeline, available at <https://github.com/kugi8412/SnakeAnalysisPhylogenomicsPipeline>. Additionally, the code and architectural framework for the Sparse Autoencoder (SAE) were adapted from my master’s thesis project, which can be accessed at <https://github.com/kugi8412/EnhancedDiscretionModel>.

The development of this project was enabled by the use of Gemini 3.1 Pro and Flash 3.5 (under a student licence). These AI tools were invaluable in helping with code generation, debugging, data processing and drafting this report.

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